

AIRSTREET · REQUIREMENTS

AirStreet requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-09-30	Amish Chadha	Status from AST-CAL-001 v0.3 after the design was made constructable (AST-DDR-003); R13 and R15 not met, proposed to Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 status from AST-CAL-001; R15 redefined as the sensor-head cost (AST-DDR-001 D1); R6, R8 and R11 rest on adopted choices
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

AirStreet requirements

These requirements are checked by calculation in [AST-CAL-001](#) v0.3. Targets remain proposals for review with the first co-design partner. On 2026-09-25 Amish accepted the recommendations in [AST-DDR-001](#) and [AST-DDR-002](#). As a result R3 is withdrawn and kept as a research question, R8, R12, R13, R15 and R16 are restated, and R15 uses the \$285 `budget_usd` . On 2026-09-30 the design was made constructable ([AST-DDR-003](#)), which raised the mass and the sensor-head cost: R13 (mass) is **not met** and is proposed to Amish with options, R15 (cost) is **over the value-engineering target** by \$18.00, five are at risk, four are met on paper, four are met by design and one is withdrawn.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (AST-CAL-001)
R1	Measure PM2.5 at street level	0 to 500 $\mu\text{g}/\text{m}^3$; hourly mean after humidity correction within $\pm 5 \mu\text{g}/\text{m}^3$ or $\pm 30 \%$ (whichever is larger) of a collocated reference	CalRig chamber check, then field collocation	At risk: within the bound up to 80 % RH; outside it above 85 % RH, where water uptake makes the correction uncertain (AST-CAL-001 E)
R2	Measure NO2 at street level	0 to 200 ppb (about 0 to 380 $\mu\text{g}/\text{m}^3$); hourly mean absolute error 5 ppb or less against a reference station after field calibration	Field collocation with a chemiluminescence reference station	At risk: hourly MAE about 3.9 ppb from a budget of literature and assumed terms (AST-CAL-001 C)
R3	Resolve NO2 near the WHO annual guideline	Withdrawn as a requirement (AST-DDR-002). Kept as a research question: can the annual mean be resolved to 2 $\mu\text{g}/\text{m}^3$ at 10 $\mu\text{g}/\text{m}^3$?	Collocation statistics (research)	Withdrawn. Annual uncertainty about 4.2 $\mu\text{g}/\text{m}^3$ (1 σ); AirStreet ranks streets and checks the EU limit (R4) and makes no WHO guideline claim
R4	Distinguish streets that exceed the EU NO2 limit value	Classify annual means above or below 40 $\mu\text{g}/\text{m}^3$ with 90 % confidence	Collocation statistics	At risk: decisive only for annual means below 34.6 or above 45.4 $\mu\text{g}/\text{m}^3$
R5	Measure temperature and humidity for corrections	$\pm 0.5 \text{ }^\circ\text{C}$ and $\pm 3 \%$ RH inside the shield; radiation error 1 $^\circ\text{C}$ or less in full sun at 1 m/s wind	CalRig chamber check; side-by-side field test against an aspirated reference	At risk: SHT45 meets the sensor figures; shield error 0.43 to 0.89 K at 1 m/s, which reads 4.2 % RH low at 85 % RH
R6	Calibrate with a stated method	T, RH and PM checked on CalRig before deployment; NO2 collocated with a reference station for at least 14 days before deployment and at least every 6 months	Calibration record per node	Met by design (AST-DDR-001 D3)
R7	Keep raw data for recalibration	Send raw NO2 working and auxiliary electrode signals, PM, T and RH with a calibration version tag	Firmware and data schema review	Met by design
R8	Report often enough for street maps	One record every 5 min through a private gateway (TwinKit by default); one every 15 min on The Things Network; hourly means published within 15 min; 7 days of store and forward	Timing, airtime and storage calculation	Met on paper: 5 min records on the private gateway; at 15 min TTN fair use is met up to SF9; 7 days in 40.3 kB
R9	Fit the FieldNode power budget	Sensor average 115 mW or less; energy neutral at the FieldNode design sun hours	Power budget calculation	Met on paper: 87.0 mW design load with the 60 s PM run; 2.33 Wh/day drawn against 7.75 Wh stored in the worst month; 6.6 days without sun
R10			Airtime calculation	

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (AST-CAL-001)
	Stay within radio duty-cycle rules	LoRaWAN airtime below 1 % in EU868 sub-bands		Met on paper: 0.082 % at SF9, 0.60 % at SF12
R11	Mount on street poles without drilling	Poles 80 to 200 mm diameter; band clamps only; inlets between 1.5 and 4 m above ground	Design review	Met by design (inlet 3.0 m, AST-DDR-001 D7; 140° V-saddles and bands cut to length fit the range, AST-DDR-003)
R12	Survive outdoors	Pod inlets face down behind insect mesh with drip lid; electronics in the FieldNode IP65 enclosure; -10 to 50 °C, with FieldNode's sun shield required at sites where ambient exceeds 45 °C	Design review, later field trial	At risk: the FieldNode sun shield is now designed (FND-DDR-003) and fixes to AirStreet's adapter plates (AST-DDR-003); SPS30 recommended range 20 to 80 % RH; condensation and insects unverified
R13	Light and compact on the pole	Mass 3.5 kg or less; frontal area 0.15 m² or less (relaxed from 0.12 m², AST-DDR-002)	Constructable model	Not met: 3.84 kg (4.00 kg with FieldNode's sun shield) against 3.5 kg; frontal area 0.143 m², met. The constructable FieldNode core and the parts that make AirStreet buildable add 0.58 kg (AST-DDR-003, A1: proposed, awaiting Amish)
R14	Collect levels only	No images, audio or personal identifiers collected or sent	Design review	Met by design
R15	Low cost	AirStreet sensor-head parts at or under the \$285 value-engineering target per node (<code>budget_usd</code> , a hypothetical control target); FieldNode core excluded and costed in FieldNode	Priced BOM (AST-CAL-001 I)	Over the value-engineering target by \$18.00: sensor head \$303.00; \$434.00 with the FieldNode core (AST-DDR-003)
R16	Serviceable in the field	Pre-allocated sensor pod exchanged in 15 min at the pole with hand tools; no sensor-level swaps at the pole (AST-DDR-002)	Design review	Met on paper: about 14 min for a pod exchange; the exchanged pod is serviced and reconnected at the reference site

■ Assumptions

- NO2 conversions use 1 ppb = 1.88 µg/m³ at 25 °C and 101.325 kPa.
- The WHO guideline level for annual NO2 is 10 µg/m³ and for PM2.5 is 5 µg/m³, as cited by the European Environment Agency (EEA, 2024). The EU annual limit value for NO2 is 40 µg/m³ under Directive 2008/50/EC, which also sets sampling inlets between 1.5 m and 4 m above ground (EUR-Lex).
- FieldNode allows 115 mW average for sensors at its 5-day autonomy ceiling, with 100 mW proposed as its published design value (FND-CAL-001); AirStreet fits either.

- `budget_usd` in `project.yaml` is \$285 for the sensor head (AST-DDR-002).
- R1 and R2 targets are proposals for review, set with reference to published field results; they are not taken from a standard.